

Miles Cochran-Branson

PHD CANDIDATE · PHYSICS

University of Washington

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Scientific machine learning, scalable inference, GPU acceleration, and statistical methods for experimental physics.

Education

University of Washington

Seattle, WA

PHD IN PHYSICS

September 2023 - present

- Courses taken: Deep Learning, Computer Systems, Quantum Field Theory, Theory of Solids
- Candidacy achieved December 2025
- Advisors: Quentin Buat, Xiangyang Ju (LBNL)

University of Washington

Seattle, WA

MASTER OF SCIENCE IN PHYSICS

September 2023 - June 2025

- Select courses taken: Quantum Mechanics, Electricity and Magnetism, Statistical Physics, Mechanics

Lawrence University

Appleton, WI

BACHELOR OF ARTS IN PHYSICS, SUMMA CUM LAUDE

September 2019 - June 2023

- Independent research in scientific machine learning and physics-informed neural networks
- Developed physics-informed neural network to solve Einstein's field equations to numerically obtain the Schwarzschild metric
- Advisors: Megan Pickett, Alexander Heaton

Technical Skills

Languages	C++, Python, Julia, Bash
Machine learning	PyTorch, ONNX, NVIDIA Triton Inference Server, multi-node distributed training on HPC
Infrastructure	Kubernetes, Docker, Dask, GPU throughput/latency profiling, Prometheus & Grafana monitoring
Scientific tools	ROOT, ATLAS Athena, NumPy/SciPy, statistical fitting and systematics, Git/GitLab CI

Research Experience

Development of Transformer-based tau reconstruction AI model

Seattle, WA and Berkeley, CA

UNIVERSITY OF WASHINGTON AND LAWRENCE BERKELEY NATIONAL LAB

Jan. 2026 - Present

- Developed and tested novel model architecture for tau reconstruction.
- Implemented custom plotting analysis suite for benchmarking and performance evaluation.

Deployment of GPU tracking as-a-Service for the ATLAS detector

Seattle, WA and Berkeley, CA

UNIVERSITY OF WASHINGTON AND LAWRENCE BERKELEY NATIONAL LAB

Jun. 2024 - Present

- Implemented custom C++ backend to run inference at scale for online data-taking.
- Developed deployment system on Kubernetes clusters for data-taking systems.
- Extensively tested throughput and latency of pipeline through custom metrics and dashboards.

Analysis of CP violation in $Z \rightarrow \tau\tau$ events

Seattle, WA and Geneva, CH

UNIVERSITY OF WASHINGTON AND CERN

Sep. 2023 - Present

- Novel analysis of di-tau final states using ATLAS data.
- Developed understanding of the novel observable, analysis framework, and statistical fitting framework.

Analysis Grand-Challenge Developer

Seattle, WA

UNIVERSITY OF WASHINGTON

Jun. 2026 - Present

- Developed and tested DASK-based tools to accelerate particle-physics analysis.
- Implemented inference as-a-Service infrastructure and statistical analysis tools.

Publications

PUBLISHED

Kondratyev, Dmitry, Benedikt Riedel, Yuan-Tang Chou, **Miles Cochran-Branson**, Noah Paladino, David Schultz, Mia Liu, Javier Duarte, Philip Harris, and Shih-Chieh Hsu. “SuperSONIC: Cloud-Native Infrastructure for ML Inference”. *PEARC ’25: Practice and Experience in Advanced Research Computing* (2025) 1–5

Contributions: Development of tools for GPU metric monitoring and deployment on test sites

The ATLAS Collaboration. “Differential cross-section measurements of Higgs boson production in the $H \rightarrow \tau^+ \tau^-$ decay channel in pp collisions at $\sqrt{s} = 13$ TeV with the ATLAS detector”. *Journal of High Energy Physics* **2025** (2025) 10 arXiv:2407.16320

Contributions: Systematics validation and implementation in data fitting

Zhao, Haoran, Yuan-Tang Chou, Yao Yao, Xiangyang Ju, Yongbin Feng, William Patrick McCormack, **Miles Cochran-Branson**, Jan-Frederik Schulte, Miaoyuan Liu, Javier Duarte, Philip Harris, Shih-Chieh Hsu, Kevin Pedro, and Nhan Tran. “Track reconstruction as a service for collider physics”. *Journal of Instrumentation* **20** (2025) P06002

Contributions: Implementation of tracking chain in ATLAS, pipeline testing and manuscript editing.

IN PREPARATION

Miles Cochran-Branson, Nicholas Kyriacou, Quentin Buat, and Ali Garabaglu. (2026). *TausRUs: a transformer-based algorithm for all tau reconstruction and identification tasks*. ATLAS PUB note: In preparation.

Contributions: Model architecture, training and optimization, and the benchmarking suite used for performance evaluation.

TECHNICAL REPORTS

Miles Cochran-Branson and Quentin Buat. (2022). *Quantifying the energy scale of hadronically decaying tau leptons at the ATLAS detector using a mixture density network*. Technical report: University of Washington Research Experience for Undergraduates (REU) program. Available online.

Contributions: Mixture density network for tau energy regression and calibration.

Awards, Fellowships, & Grants

2026	Office of Science Student Research (SCGSR) Fellowship, U.S. Department of Energy	\$ 21,600
	Analysis Fellowship, IRIS-HEP	\$ 10,000
2025	Graduate Research Fellowship Program, Honorable Mention, National Science Foundation	
2024	Western Advanced Training for Computational High-Energy Physics (WATCHEP) Fellowship, Department of Energy (DOE)	\$ 65,000 / year
2023	Provost Award, University of Washington	\$ 10,000
	Physics Department Fellowship, University of Washington	\$ 5,000
2022	J. Bruce Brackenridge Prize for excellence in physics, Lawrence University	
	Maurice Cunningham Phi Beta Kappa Prize for highest GPA in junior class, Lawrence University	
2021	Sir Isaac Newton (SIN) award for creativity in computational physics problem-solving, Lawrence University	
	Ralph White Prize in Mathematics, Lawrence University	

Select Presentations

INVITED TALKS

Miles Cochran-Branson. (2025). *Standard Model Physics in $\tau\tau$ final states*. Invited talk: ATLAS Standard Model Workshop, University of Science and Technology of China (USTC), Hefei, China.

CONTRIBUTED TALKS

- Miles Cochran-Branson**, Quentin Buat, Nicholas Kyriacou, and Ali Garabaglu. (2026). *Developing a machine-learning algorithm for all tau reconstruction and identification tasks*. Oral presentation: Tau Combined Performance Group Workshop, CERN, CH.
- Miles Cochran-Branson**, Xiangyang Ju, and Yuan-Tang Chou. (2026). *GPU-Accelerated Particle Tracking as-a-Service*. Oral presentation: Fast Machine Learning for Science Conference, UC San Diego, San Diego, CA.
- Miles Cochran-Branson**. (2025). *R_{QCD} fake estimation in $Z \rightarrow \tau\tau$ spin measurement*. Oral presentation: Tau Combined Performance Group Workshop, CERN, Switzerland.
- Miles Cochran-Branson** and Xiangyang Ju. (2025). *Integrating GNN4ITk into GPU tracking pipelines*. Oral presentation: EF-Tracking Workshop, Chateau de Bossey, Switzerland.
- Miles Cochran-Branson**, Quentin Buat, Matt Foresi, and Chris Young. (2024). *Search for CP violation in the $Z \rightarrow \tau\tau$ channel*. Oral presentation: US-ATLAS Annual Meeting, University of Washington, Seattle, WA.
- Miles Cochran-Branson**, Xiangyang Ju, Yuan-Tang Chou, and Haoran Zhao. (2024). *GPU-Accelerated Particle Tracking as-a-Service*. Oral presentation: US LHC Users Association Annual Meeting, SLAC National Accelerator Laboratory, Menlo Park, CA.
- Miles Cochran-Branson**, Xiangyang Ju, Yuan-Tang Chou, and Haoran Zhao. (2024). *Implementation of *traccc* as-a-Service*. Oral presentation: A3D3 All-Hands Meeting and Fast Machine Learning Conference, Purdue University, Lafayette, IN.

Outreach & Professional Development

SERVICE AND OUTREACH

- 2026 **AI Future Lab Lecturer and Lab Leader**, Prepared lab exercises on introductory AI coding concepts for 30 high-school students from Taiwan, and gave a lecture on AI ethics.
- 2025 **U.W. A3D3 Machine Learning Hackathon**, Organized an ML hackathon at University of Washington for more than 30 students on AI prediction tasks using scientific data.
- 2025 **U.S. LHC Users Association D.C. Trip**, High-Energy Physics funding ambassador, advocating for funding in Congress directly with House and Senate members.
- 2024 **Exploring the Quantum Universe with Artificial Intelligence**, Undergraduate Symposium Moderator and Mentor.
- 2024 **IMOD outreach with Rainier Prep. Middle School**, Introduced experimental science to 90 fifth grade students through engaging interactive activities.

DEVELOPMENT

- WATCHEP Summer School**. *Lawrence Berkeley National Lab, Summer 2025*. This joint summer school with other computing physics groups in cosmology and particle physics focused on the breadth of physics and computing currently being explored in science and industry.
- Deep Learning at Scale Training**. *NERSC facility at Lawrence Berkeley National Lab, 2025*. This training focused on distributed training and data parallelization techniques for training models on high-performance computing systems.
- Machine Learning for Fundamental Physics School**. *Lawrence Berkeley National Lab, Summer 2024*. This workshop focused on tools to deploy machine learning models for a variety of computing needs. Most relevant topics included deployment of models on FPGAs, Differential Programming, Transformers, and Unfolding using machine learning.

MENTORING

As a PhD student I have mentored four undergraduates in research techniques by introducing the ATLAS experiment to them, as well as instilling an excitement for physics research and discovery. In this role I organize weekly meetings and check-ins, as well as field questions and provide academic guidance.

MEMBERSHIPS

- Accelerated AI Algorithms for Data-Driven Discovery (A3D3)
- Phi Beta Kappa (National Honors Society)
- Sigma Pi Sigma (Physics Honors Society)